Class Verilog Standards

CSEE 4280

What you see in class

- Examples and lectures will show many different, valid Verilog:
  - Syntax
  - Methods
  - Behavioral only
  - Synthesizable
  - Test bench
- We need to pick a class set of standards for your project Verilog.

Class Verilog Standards

I will use these as part of grading standards.
This is a dynamic list.

We will use Ansi instead on Non-Ansi for inputs/outputs

Non-Ansi

```
module myModule (a, b, c);
        input a;
        input int b;
        output signed logic [7:0] c;
        // ... module body ...
  endmodule
```

Ansi

```
module myModule (
        input a,
        input int b,
        output signed logic [7:0] c
);
  // ... module body ...
endmodule
```

Naming Conventions

- Active Low
  - n_ or _n (I tend to use the second)
- Flops
  - _r or _ff (I prefer the second. If we were using latches, _l)
  - Optional for vector of flops: _v
  - Optional for array of flops: _a
- Clock
  - clk or clock
  - If part of a name _clk

Design Conventions

- Data busses are separate from control
- Blocking and non-blocking are separate
  - No combinational in the <= non-blocking always unless using for specific FF.
  - Period!
- Do not use for loops in synthesis. Although Vivado supports it, it is difficult and usually better ways to do this.
  - For this class you may use with behavior and test benches
  - You may not use for synthesis without my permission

Design Conventions

- Do not gate clocks
- All FF-like items must have a reset value.
- Mux FF inputs to hold prior values
- Always use a default: for last compare value of case statement.
- Always use a last else as a default in if – else conditionals

System Conventions

- Meaningful, descriptive comments
  - Meaningful variable, flop, I/O names
  - Top of file
  - Top of block
  - Top of major always
  - Top of major case
  - Top of major if-then-else
- No combinational logic at top level of hierarchical designs.
- All designs must have a corresponding testbench
- The testbench must be self-checking